RESEARCH PROPOSAL

Leveraging Graph Structures for Lightweight Multi-Person Pose Tracking

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Assignment 3
Introduction to Research EIN 732
Department of Electrical, Electronic and Computer Engineering
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Category	Assignment Element		Comment	Maximum	Supervisor	Lecturer	External	Final Mark
1. Title			Is the title concise?					
2. Abstract	What was found in the body of science; what is proposed; what is expected; significance.		Does the abstract contain relevant information?					
				5	4			
3. Introduction	Describe the body of science that underpins the research proposal.		Are the statements in the introduction logically developed? (Lose up to 50%.)					
	Define the problem to be addressed (Identify the research gap) and place it in the context of the body of science.							
				10	8			
4. The Research Proposal	Describe the Research question(s) that you will address.		Is the Research Proposal logically developed? (Lose up to 50%.)					
	Formulate your Hypothesis.							
	Describe the research procedure you intend following. (Evaluate the proposed procedure in terms of good argument, etc.)							
				20	15			
5. Conclusion	Evaluate the expected outcome of your work and your contribution;							
	the importance in the broader scientific context;							
	state the main issues around your topic.							
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Leveraging Graph Structures for Lightweight Multi-Person Pose Tracking

Dean Mills

Abstract

This report proposes a novel approach to decouple pose estimation from pose tracking, aiming to create a lightweight and efficient pose tracking system. The proposed method leverages a Graph Attention Network (GAT) and Deep Embedded Clustering (DEC) to effectively group joints into individual humans. By incorporating joints from previous frames into a single large graph, the GAT utilizes message passing and attention mechanisms to extract spatial and temporal information. This approach introduces pseudo-physical constraints and emphasizes important features, contributing to a reduced feature space. The resultant feature space reduction facilitates the use of shallower networks, which aligns with the inherent properties of graph-based networks. This decoupled tracking system aims to decrease the computational overhead of pose estimation tasks by producing a more efficient tracking system.

Index Terms

Pose Estimation, Pose Tracking, Graph Attention Networks, Multi-Person Pose Tracking, Deep Embedded Clustering

I. Introduction

Pose estimation is the process of identifying the configuration of a person's body parts, such as the position of the head, limbs, and joints, within an image or video. This field has been an active area of research for many years, with methodologies evolving significantly over time. Pose estimation techniques can be broadly categorized into two types: top-down and bottom-up. Top-down methods first detect the entire human body and then estimate the pose within the detected region, while bottom-up methods first detect individual joints and then group them to form complete poses.

Several classical approaches have been proposed for pose estimation, each with its own strengths and limitations. Notable methods include Relevance Vector Machines [1], Mixture of Experts [2], Locality-Constrained Linear Coding [3], Relevant Component Analysis [4], Bag of Words Based Methods [5], Random Forests [6], Hough Forests [7], Pixel or Superpixel-

Based Methods [8], and Motion-Based Models [9]. These older methods served their purpose but faced limitations as the complexity and requirements of pose estimation tasks grew.

As the need to handle more sophisticated scenarios such as 3D pose estimation and multi-person detection arose, deep learning-based models became increasingly prominent. One of the earliest and most influential deep learning models for pose estimation was DeepPose [10], which utilized AlexNet [11] as its backbone. Backbone architectures refer to the core convolutional neural network used to extract features from the input images, forming the basis for the pose estimation task. Various backbone architectures have been employed in different models, including R-CNN [12], Fast R-CNN [13], Faster R-CNN [14], Feature Pyramid Networks (FPN) [15] and Mask R-CNN [16].

The development of the VGG network [17], which enabled the construction of deeper networks due to its use of small convolutional filters, marked a significant advancement in pose estimation. The small filters (3x3) used in VGG allowed for a substantial increase in the depth of the network without a corresponding increase in the number of parameters, facilitating more complex and detailed feature extraction.

VGG's architecture was employed in several important pose estimation models, such as Convolutional Pose Machines (CPM) [18] and, notably, OpenPose [19]. OpenPose utilizes the first 19 layers of VGG for feature extraction and introduces the concept of part affinity fields. These fields are vector fields that encode the location and orientation of limbs, effectively learning the associations between detected joints. By generating part affinity fields, OpenPose can robustly group detected joints into individual human poses, even in complex, multi-person scenarios.

ResNet [20] has emerged as the predominant backbone architecture due to its ability to construct very deep networks thanks to residual connections. These connections help mitigate the vanishing gradient problem, allowing for more effective training of deeper networks. The complexity of pose estimation tasks often necessitates such deep networks, but it should be a point of concern that increasingly deeper networks are being used. As networks become deeper, they require more computational resources and can become less efficient, making them less practical for real-time applications and deployment on devices with limited processing power. Additionally, the increased complexity can lead to overfitting and greater difficulty in training and tuning the models. Notable pose estimation models using ResNet include Cascade Feature Aggregation for Human Pose Estimation [21], Stacked Hourglass Networks for Human Pose Estimation [22], Simple Baselines for Human Pose Estimation and Tracking [23], Cascaded Pyramid Network for Multi-Person Pose Estimation [24], and High-Resolution

Network for Human Pose Estimation [25], [26].

Due to the requirements of multi-person tracking, pose estimation has evolved to encompass both pose estimation and pose tracking. Pose tracking, which is the focus of this report, involves mapping detected joints to individual humans. OpenPose, for instance, incorporates part affinity fields to facilitate this mapping, adding to its complexity. Pose tracking is particularly challenging due to occlusions and ambiguities. Occlusions occur when parts of the body are hidden from view, either by other body parts or by other individuals in the scene. Ambiguities arise from the close proximity of similar joints from different individuals, making it difficult to accurately associate each joint with the correct person. These challenges make pose tracking a critical area of research within the broader field of pose estimation.

II. Graphs

Graphs are a fundamental concept in this proposal as they provide a powerful framework for representing and processing complex relationships between entities. In general, a graph is a mathematical structure consisting of a set of nodes (or vertices) and a set of edges (or links) that connect pairs of nodes. This structure is particularly useful for modeling various types of data, including social networks, biological networks, and, in this context, human joints in pose estimation.

A. General Concept of Graphs

A graph G is defined as $G = (V, E)$, where:

- V is a set of nodes.
- E is a set of edges, where each edge $e \in E$ connects a pair of nodes (u, v) with $u, v \in V$.

Graphs can be directed or undirected. In a directed graph, edges have a direction, indicating the relationship goes from one node to another. In an undirected graph, edges have no direction, representing mutual relationships.

B. Graph Neural Networks

Graph Neural Networks (GNNs) [27] are a class of neural networks designed to operate on graph-structured data. GNNs generalize the concept of convolutional neural networks (CNNs) to graphs, allowing them to capture dependencies and relationships between nodes. The core idea of GNNs is to aggregate information from a node's neighbors to update its representation iteratively. This process is known as message passing.

The message passing framework for a GNN can be summarized as follows:

- **Initialization**: Each node v is initialized with a feature vector $h_v^{(0)}$.

- **Message Passing**: For each iteration t, each node aggregates information from its neighbors:

$$m_v^{(t+1)} = \text{AGGREGATE}^{(t)}\left(\{h_u^{(t)} : u \in \mathcal{N}(v)\}\right) \tag{1}$$

- **Update**: The node updates its feature vector using the aggregated information:

$$h_v^{(t+1)} = \text{UPDATE}^{(t)}(h_v^{(t)}, m_v^{(t+1)}) \tag{2}$$

The AGGREGATE and UPDATE functions can vary, but common choices include sum, mean, or max pooling for aggregation and neural network layers for updates.

C. Graph Convolutional Networks

Graph Convolutional Networks (GCNs) [28] are a specific type of GNN that perform convolution operations on graphs. GCNs extend the concept of convolution from grid-like data structures (such as images) to graph structures. The key idea is to apply a convolutional filter over the graph to aggregate features from neighboring nodes.

A typical GCN layer can be defined as:

$$H^{(t+1)} = \sigma\left(\tilde{D}^{-1/2}\tilde{A}\tilde{D}^{-1/2}H^{(t)}W^{(t)}\right) \tag{3}$$

where:

- $\tilde{A} = A + I$ is the adjacency matrix with added self-loops.
- $\tilde{D}$ is the degree matrix of $\tilde{A}$.
- $H^{(t)}$ is the node feature matrix at layer t.
- $W^{(t)}$ is the weight matrix for layer t.
- σ is an activation function, such as ReLU.

GCNs effectively aggregate information from a node's local neighborhood, enabling the network to learn node representations based on their structural context in the graph.

D. Graph Attention Networks

Graph Attention Networks (GATs) extend Graph Neural Networks (GNNs) by incorporating an attention mechanism to dynamically weight the importance of neighboring nodes during the aggregation process. This allows GATs to focus on the most relevant neighbors, improving the representation learning process. The concept of attention has its roots in transformers [29], which are used extensively in large language models [30]. In natural language processing, attention mechanisms help the model identify which words in a sentence are more important than others, such as assigning more weight to nouns and verbs over prepositions, as these often carry more critical information.

The key steps in a GAT layer are:

- **Self-Attention**: Compute attention coefficients for each pair of connected nodes (u, v):

$$e_{uv} = \text{LeakyReLU}\left(\mathbf{a}^T \left[Wh_u \,||\, Wh_v\right]\right) \tag{4}$$

  where $\mathbf{a}$ is a learnable weight vector, W is a shared weight matrix, and $||$ denotes concatenation.

- **Normalization**: Normalize the attention coefficients using the softmax function:

$$\alpha_{uv} = \frac{\exp(e_{uv})}{\sum_{k \in \mathcal{N}(v)} \exp(e_{vk})} \tag{5}$$

- **Aggregation**: Compute the new node features as a weighted sum of the neighbors' features:

$$h'_v = \sigma\left(\sum_{u \in \mathcal{N}(v)} \alpha_{uv} W h_u\right) \tag{6}$$

In the context of pose estimation, attention mechanisms can similarly prioritize certain types of information. For example, it may be found that the spatial position of a person's hip is a particularly valuable feature for accurately tracking poses. By dynamically weighting the importance of such features, GATs can better model the complex relationships between joints, thereby enhancing the overall accuracy and robustness of the pose tracking system. This approach allows the network to focus on the most informative parts of the graph, making it more efficient in handling the complexities of multi-person pose estimation and tracking.

III. Research Questions

The following research questions are used as a baseline to explore and address the challenges in multi-person pose tracking. These questions aim to guide the investigation into effective methodologies for improving the accuracy, efficiency, and robustness of pose tracking systems.

1) Is it viable to decouple pose estimation and pose tracking?
2) Is there a way to reduce the training time for pose tracking tasks?
3) How can physical constraints be incorporated into pose tracking models?
4) How can temporal consistency be maintained in pose tracking across multiple frames?
5) How can lightweight models be designed without sacrificing pose tracking accuracy?

IV. Hypothesis

Modeling joint probabilities as a graph then using GAT and incorporating historic data (previous frames) will allow for a significant reduction in feature space. This reduction

is achieved through the GAT's message passing and attention mechanisms, which extract important spatial information and dynamically weight features on joints, such as spatial coordinates. Additionally, by including historic data, the GAT gains a better understanding of how joints move over time, effectively creating a pseudo-physics-based system. This temporal context allows the network to infer movement patterns and constraints, providing more information than analyzing frames in isolation. The processed feature space will then be effectively clustered using Deep Embedded Clustering (DEC) [31] to group joints associated with individual humans.

Explanation: The key components of this hypothesis are:

1) **Graph Attention Network (GAT)**:
   - **Message Passing**: GATs facilitate the exchange of information between nodes (joints), allowing for the aggregation of spatial dependencies and relationships.
   - **Attention Mechanism**: By dynamically weighting the importance of different nodes, GATs can prioritize relevant features and enhance the representation of critical joints.
2) **Historic Data**:
   - **Temporal Context**: Incorporating data from previous frames enables the model to learn and enforce temporal consistency, capturing how joints move over time and improving tracking accuracy.
3) **Feature Space Reduction**:
   - **Dimensionality Reduction**: The combined effect of message passing and attention in GATs leads to a more compact and informative feature space, reducing the complexity of the data without losing essential information. Additionally, incorporating historic data introduces movement constraints, which further contributes to dimensionality reduction. By understanding the temporal dynamics of joint movements, the GAT can effectively filter out irrelevant variations and focus on the most significant features. This dual approach ensures that the feature space is not only reduced in size but also enriched with meaningful spatial and temporal information.
4) **Deep Embedded Clustering**:
   - **Effective Grouping**: The reduced and enriched feature space is then processed using DEC, which groups joints into clusters corresponding to individual humans. This step enhances the efficiency and accuracy of multi-person pose tracking.

This hypothesis sets the foundation for the proposed research, emphasizing the potential benefits of combining GATs, historic data, and DEC for enhanced pose tracking in multi-person scenarios.

V. Approach

The proposed approach for solving the problem of multi-person pose tracking involves several key steps. These steps include extracting joint information from images, constructing a graph representation, processing this graph through a GAT, clustering the features using DEC, and performing a sanity check to ensure the accuracy of the tracked poses. The final step involves visually representing the tracked poses by drawing lines connecting the joints of each individual.

A. Joint Extraction from Images

The first step involves extracting the joint coordinates from the input images. This will be achieved using OpenPose, a widely-used pose estimation framework. OpenPose detects keypoints (joints) of human bodies in images and provides their coordinates along with confidence scores. The framework can detect multiple keypoints per person, including joints such as elbows, knees, shoulders, and ankles. The output from OpenPose will serve as the initial input for further processing, providing a comprehensive set of joint coordinates and associated confidence scores for each detected person in the frames.

B. Graph Construction

After obtaining the joint coordinates, a graph is constructed where each node represents a joint. The graph includes nodes corresponding to joints detected in the last few frames of video. To make the method frame rate agnostic, the graph can be constructed from the joints detected in the last n seconds. This ensures that the graph maintains temporal consistency regardless of the frame rate. Edges are added between nodes based on spatial proximity within a frame (connecting joints that are anatomically close) and temporal continuity across frames (connecting the same joint across consecutive frames). This step ensures that both spatial and temporal relationships between joints are captured in the graph.

C. Graph Attention Network Processing

The constructed graph is then processed through a GAT. The GAT allows for message passing between nodes, enabling the network to aggregate information from neighbouring nodes. Each node's feature vector, initially representing joint coordinates, confidence, and

joint type, is updated based on its neighbour's features. The attention mechanism within the GAT dynamically weights the importance of each feature on neighboring nodes, allowing the network to prioritize relevant features and improve the representation of each joint. This helps the network capture significant spatial and temporal dependencies, resulting in a set of enriched feature vectors for each node. These enriched features facilitate more accurate and efficient pose tracking by effectively modeling the complex relationships and movements of joints.

D. Deep Embedded Clustering

Once the features have been extracted by the GAT, DEC is applied to group the joints into clusters corresponding to individual humans. The clustering algorithm operates on the reduced feature space, which has been compacted through the GAT's processing. By embedding the high-dimensional joint features into a lower-dimensional space, clustering algorithms like K-means or DBSCAN can more effectively identify and group related joints. This step ensures that joints are accurately associated with their respective individuals, even in complex multi-person scenarios, improving both efficiency and accuracy of the tracking system.

E. Sanity Check

After clustering, a sanity check is performed to verify the accuracy of the tracked poses. This involves ensuring that each individual has a consistent set of joints, without any anomalies such as a single human having two right elbows or other anatomical inconsistencies. The sanity check uses predefined rules based on human anatomy to validate the clusters and correct any possible errors from the clustering process. This step helps in maintaining the integrity of the tracked poses and ensures that the resulting clusters correspond to biologically plausible human skeletons.

F. Visual Representation

The final step involves creating a visual representation of the tracked poses. Lines are drawn connecting the joints of each individual, forming a coherent skeleton structure. This visual representation aids in verifying the accuracy of the tracking and provides an intuitive understanding of the results. The visualized poses can be overlaid on the original images or displayed separately for analysis. This step not only helps in validating the results but also provides a clear and interpretable way to present the findings, making it easier to communicate the effectiveness of the proposed method.

VI. Discussion

The proposed approach of using GATs with historic data and DEC for multi-person pose tracking introduces several interesting points for discussion. This section elaborates on the anticipated benefits and potential alternatives to the chosen methods.

One of the fundamental assumptions in this approach is that spatial information will help in accurately grouping joints belonging to the same individual. For instance, if a wrist moves significantly farther relative to the hip within a short time frame, it is likely that this wrist does not belong to the same person. By leveraging GATs, the network can capture such spatial relationships and dynamically weight the importance of each joint's position relative to others. This mechanism helps in ensuring that joints exhibiting coherent movement patterns are grouped together, thereby improving the accuracy of pose tracking.

Incorporating historic data from previous frames introduces implicit physical constraints into the model. Although the approach does not explicitly simulate physics, it benefits from the fact that body parts can only move within reasonable limits over a short period. This constraint helps in disambiguating joints that might otherwise be difficult to track, especially in crowded scenes. For example, if two right wrists are initially close together, and one wrist moves disproportionately, the model can infer that this joint likely belongs to a different person, potentially occluded in the previous frame. This temporal consistency is crucial for maintaining accurate tracking over time.

While GATs offer the advantage of an attention mechanism that can prioritize certain joints based on their importance, it is worth considering if a GNNs/GCNs might suffice for this task. GNNs/GCNs also facilitate message passing and can capture the spatial relationships between joints. The primary difference is that GNNs/GCNs do not inherently include an attention mechanism. For instance, the x/y coordinate of the neck might be a critical feature, and GATs can dynamically assign higher importance to such features. However, if the computational complexity of GATs becomes a bottleneck, GNNs/GCNs could be a viable alternative, providing a simpler yet effective approach for message passing without the overhead of attention calculations.

DEC is chosen for its ability to make the clustering process differentiable, allowing for end-to-end training and easier backpropagation of errors. However, this approach might introduce additional complexity and slow down the processing. Classical clustering methods such as DBSCAN or K-means could be considered as simpler alternatives. These methods are well-established and can effectively cluster joint features based on spatial proximity. The trade-off

here is between the elegance and potential performance benefits of DEC versus the simplicity and speed of classical methods. DEC's differentiability makes it an attractive option for integrating clustering into the learning process, but it is essential to evaluate whether the additional computational cost is justified by the gains in accuracy and robustness.

Graph networks, are typically not very deep compared to other types of neural networks like CNNs. This characteristic is often due to the phenomenon of over-smoothing, where increasing the number of layers can cause node features to become indistinguishably similar. Over-smoothing can degrade the performance of graph networks, making it challenging to maintain distinct feature representations across nodes. Thus, practical implementations of graph networks often limit the depth to avoid this issue and ensure that the learned features remain informative and discriminative.

VII. Conclusion

The proposed solution for multi-person pose tracking leverages GATs, historic data, and DEC. This method extracts joint coordinates from images using OpenPose, constructs a graph representation from the joint data across multiple frames, and processes this graph through a GAT to capture spatial and temporal dependencies. By incorporating historic data, the model enforces implicit physical constraints, enhancing tracking accuracy in dynamic scenes. The chosen method aims to create a lightweight tracking system for multi-person scenarios that can be applied independently of the underlying pose estimation method. This approach has the potential to offer a flexible and efficient solution for pose tracking, significantly contributing to the broader scientific community by reducing the complexity, depth, and therefore the training time and speed of execution, of pose estimation and tracking methods.

References

[1] M. E. Tipping, "Sparse bayesian learning and the relevance vector machine," *Journal of machine learning research*, vol. 1, no. Jun, pp. 211–244, 2001.

[2] L. Sigal and M. J. Black, "Predicting 3d people from 2d pictures," in *International Conference on Articulated Motion and Deformable Objects*. Springer, 2006, pp. 185–195.

[3] J. Wang, J. Yang, K. Yu, F. Lv, T. Huang, and Y. Gong, "Locality-constrained linear coding for image classification," in *2010 IEEE computer society conference on computer vision and pattern recognition*. IEEE, 2010, pp. 3360–3367.

[4] A. Kanaujia, C. Sminchisescu, and D. Metaxas, "Semi-supervised hierarchical models for 3d human pose reconstruction," in *2007 IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 2007, pp. 1–8.

[5] H. Ning, W. Xu, Y. Gong, and T. Huang, "Discriminative learning of visual words for 3d human pose estimation," in *2008 IEEE Conference on Computer Vision and Pattern Recognition*. IEEE, 2008, pp. 1–8.

[6] J. Y. Chang and S. W. Nam, "Fast random-forest-based human pose estimation using a multi-scale and cascade approach," *ETRI Journal*, vol. 35, no. 6, pp. 949–959, 2013.

[7] R. Girshick, J. Shotton, P. Kohli, A. Criminisi, and A. Fitzgibbon, "Efficient regression of general-activity human poses from depth images," in *2011 International Conference on Computer Vision*. IEEE, 2011, pp. 415–422.

[8] H. Wang and D. Koller, "Multi-level inference by relaxed dual decomposition for human pose segmentation," in *CVPR 2011*. IEEE, 2011, pp. 2433–2440.

[9] B. Sapp and B. Taskar, "Modec: Multimodal decomposable models for human pose estimation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2013, pp. 3674–3681.

[10] A. Toshev and C. Szegedy, "Deeppose: Human pose estimation via deep neural networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 1653–1660.

[11] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," in *Advances in neural information processing systems*, 2012, pp. 1097–1105.

[12] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 580–587.

[13] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 1440–1448.

[14] S. Ren, K. He, R. Girshick, and J. Sun, "Faster r-cnn: Towards real-time object detection with region proposal networks," *Advances in neural information processing systems*, vol. 28, 2015.

[15] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 2117–2125.

[16] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2017, pp. 2961–2969.

[17] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.

[18] S.-E. Wei, V. Ramakrishna, T. Kanade, and Y. Sheikh, "Convolutional pose machines," in *Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*, 2016, pp. 4724–4732.

[19] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, "Realtime multi-person 2d pose estimation using part affinity fields," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 7291–7299.

[20] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[21] Z. Su, M. Ye, G. Zhang, L. Dai, and J. Sheng, "Cascade feature aggregation for human pose estimation," *arXiv preprint arXiv:1902.07837*, 2019.

[22] A. Newell, K. Yang, and J. Deng, "Stacked hourglass networks for human pose estimation," in *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11-14, 2016, Proceedings, Part VIII 14*. Springer, 2016, pp. 483–499.

[23] B. Xiao, H. Wu, and Y. Wei, "Simple baselines for human pose estimation and tracking," in *Proceedings of the European conference on computer vision (ECCV)*, 2018, pp. 466–481.

[24] Y. Chen, Z. Wang, Y. Peng, Z. Zhang, G. Yu, and J. Sun, "Cascaded pyramid network for multi-person pose estimation," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 7103–7112.

[25] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 5693–5703.

[26] Z. Zhang, J. Tang, and G. Wu, "Simple and lightweight human pose estimation," *arXiv preprint arXiv:1911.10346*, 2019.

[27] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE transactions on neural networks*, vol. 20, no. 1, pp. 61–80, 2008.

[28] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," *arXiv preprint arXiv:1609.02907*, 2016.

[29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

[30] W. X. Zhao, K. Zhou, J. Li, T. Tang, X. Wang, Y. Hou, Y. Min, B. Zhang, J. Zhang, Z. Dong *et al.*, "A survey of large language models," *arXiv preprint arXiv:2303.18223*, 2023.

[31] J. Xie, R. Girshick, and A. Farhadi, "Unsupervised deep embedding for clustering analysis," in *International conference on machine learning*. PMLR, 2016, pp. 478–487.